IN PRESS, PSYCHOLOGICAL RESEARCH

How general is ensemble perception?

Ting-Yun Chang[1], Oakyoon Cha[2], Rankin McGugin[1], Andrew Tomarken[1] and Isabel Gauthier[1]

[1]Department of Psychology, Vanderbilt University, Nashville, TN, USA
[2]Department of Psychology, Sungshin Women's University, Seoul, South Korea

Abstract: 250 words **Main text:** 9087 words **References:** 45
Figures: 3 **Tables**: 3

Funding
This work was supported by the David K. Wilson Chair Research Fund from Vanderbilt University awarded to Isabel Gauthier and the 2021-2022 Taiwanese Overseas Pioneers Grants (TOP Grants) for PhD Candidates from Ministry of Science and Technology, Taiwan, awarded to Ting-Yun Chang.

Correspondence **during the review process** concerning this article should be addressed to Isabel Gauthier, Vanderbilt University, 111 21st Avenue South, Nashville, TN 37240, USA.
Email: Isabel.gauthier@vanderbilt.edu, phone: 615 494 2944, No fax.
ORCID: 0000-0002-6249-4769

Correspondence **after the review process** should be addressed to Ting-Yun Chang, Department of Psychology, Vanderbilt University, 111 21st Avenue South, Nashville, TN 37240, USA.
E-mail: ting-yun.chang@vanderbilt.edu, phone: 615 710-0807, No fax.
ORCID: 0000-0002-6582-6482

Conflict of Interest statement
On behalf of all authors, the corresponding author states that there is no conflict of interest.

Data availability statements
Data are available on figshare: https://doi.org/10.6084/m9.figshare.21875355.v1

Abstract

People can summarize features of groups of objects (e.g., the mean size of apples). Claims of dissociations or common mechanisms supporting such ensemble perception (EP) judgments have generally been made on the basis of correlations between pairs of tasks. These correlations can be inflated because they use the same stimuli, summary statistics and/or task format. Performance on EP tasks also correlates with that on object recognition (OR) tasks. Here, we seek evidence for a general EP ability that is also distinct from OR ability. Two-hundred participants completed three tasks that did not overlap in stimuli, summary statistic or task format. Participants performed a diversity comparison for arrays of nonsense blobs, a mean identity judgment with ensembles of Transformer toys, and the novel object memory task with novel objects (NOMT-Greeble). We hypothesized that *EP* contributes to the first two of these tasks, while *OR* contributes only to the last two. Performance on the two tasks suggested to tap an EP ability were correlated after controlling for the third task. Confirmatory factor analysis was used to test our predictions without the confound of measurement error. Correlations between factors assumed to share influence from EP or from OR was higher than that between the factors that we expect did not share these influences. The results provide the first clear evidence for a domain-general EP ability distinct from OR. We argue that understanding such a general ability will require a change in designs and analytical approaches in the study if individual differences in EP.

Keywords: ensemble perception, individual differences, object recognition, summary statistics

1. Introduction

When you choose the box of assorted chocolates that best suits your preferences, determine if your audience is offended or amused by your joke, or change lanes on highway based on the average speed of cars, you base visual decisions on summary statistics for groups of objects. Judgments such as these have been studied under the broad rubric of Ensemble Perception (EP; Ariely, 2001). People can make a host of EP judgments, estimating different summary statistics (e.g., mean, variance, range, numerosity) for a wide range of visual features (e.g., size, orientation, color, contrast, motion direction, for review see Whitney and Yamanashi Leib, 2018). For each of the these tasks, there is often evidence for reliable individual differences, which may or may not correlate with performance in the recognition of individual objects depending on the tasks (e.g., (Haberman et al., 2015). However, using EP to describe these different tasks does not guarantee that they tap the same mechanisms. Haberman et al. (2015) were the first to ask whether EP judgments may be supported by a general "ensemble processor" regardless of the visual domain. We address the same question here, using an approach that seeks evidence for an "ensemble processor" that could generalize not only across domains, but also across tasks and different summary statistics.

If there is a common ability that contribute to performance on all EP tasks, it could predict performance across a variety of situations where EP has been suggested to apply. This may include tasks as different as texture processing, interpreting climate maps, making fast buy/sell decisions about stocks, estimating the speed of cars in traffic, or screening mammograms (e.g., Balas et al., 2009; Brennan et al., 2018). If there is no general EP ability, the underlying construct proposed to link performance on these tasks would be lacking. It would also mean that theoretical efforts to explain EP (Brady & Alvarez, 2015; Harrison et al., 2021) would need to explain why reliable individual differences in different EP tasks are not related to each other, and offer models for each distinct EP task. Given the range of tasks that are studied under the EP umbrella, there may be more than one higher-level EP ability that can account for performance across different task subsets (Haberman et al., 2015). Studying the structure of EP abilities can facilitate the development of process models, by clarifying how many different mechanisms are needed to explain these judgments. Efforts in modeling cognition (e.g, Schubert et al., 2022) are starting to address challenges presented by individual differences. But modeling of cognitive processes is often task-specific and can show limited generalization across paradigms (Logan, 1988). There are new efforts in modeling mechanisms for EP (Jeong & Palmeri, 2023; Robinson & Brady, 2022; Utochkin et al., 2023) but they are not models that are sufficiently general to account for all EP tasks or to make predictions about the structure of EP abilities.

The main approach to test for common abilities is rooted in studying correlations of performance across tasks. To "carve nature at its joints," individual differences can be used to identify remote associations between dissimilar tests and proximal dissociations between similar tests. Such results can provide useful landmarks to begin to explore the structure of abilities, especially in a space as large as that of EP, where it is difficult to imagine multivariate studies that would include all possible combinations of task and stimulus characteristics. We hope to provide a proof of concept for evidence of a general EP ability, in a manner that we hope can shift the approach used in this field. We limit ourselves here to explicit EP tasks because recent evidence suggests that abilities relevant to explicit EP judgments may differ from those that influence performance in implicit EP tasks (Hansmann-Roth et al., 2021). There is a growing body of work measuring correlations in performance across EP tasks, suggesting both associations and dissociations. We propose below that evidence of dissociations has often been relatively weak (and does not always replicate), that evidence for associations may often be explained by superficial test characteristics, and that a convincing test of a higher-order EP ability has been lacking because of the kinds of pairwise comparisons that are tested.

1.1. Weak dissociations

A number of proximal dissociations have been suggested between performance on relatively similar EP tasks, but later studies with more power found evidence of association instead. One kind of dissociation concerns different types of summary statistics (e.g., average and variance, or numerosity and average). Several studies (e.g., Khvostov & Utochkin, 2019; Yang et al., 2018) concluded that judgments for different statistical properties do not correlate, suggesting that the processing of each statistic happens in a different module. These studies generally had low power (N < 30). Using a Bayesian approach in which data were collected until evidence favored either a correlation or no correlation, recent work measured individual differences in judgments for: i) the size of a single circle, ii) the variance in size within an array of circles, and iii) the average size within an array of circles (Cha et al., 2022). After controlling for performance with single circles, the average and variance judgments were correlated (r = .31). Another dissociation was proposed for the mechanisms supporting EP judgments regarding the same summary statistic but for different features. This was suggested on the basis of the absence of a correlation between judgments for the average length and orientation of arrays of lines (Yörük & Boduroglu, 2020). This work reported three studies that were underpowered; some correlations reached values approaching .3, yet there was only sufficient power to detect much larger effects. More recent work collected data until evidence favored either a correlation or no correlation and addressed design problems that could have dampened the effect size (e.g., very short lines that may limit orientation judgments, a wide range of orientations that may complicate the use of a linear mean as an estimate of circular mean). In two studies, Kacin et al. (2021) found decisive support for a positive correlation between these two low-level judgments (around r = .6 across two studies). These results illustrate the importance of power and precision, especially when data suggest null effects. Despite this, we acknowledge that the original null effects may have been particularly intriguing, since correlations between these tasks may be explained by the commonality in task format, stimuli, or summary statistics. In addition, the focus has primarily been on the absence vs. presence of a correlation, rather than on the precision of correlation estimates.

1.2. Weak associations

A general challenge that can be raised about prior work on individual differences in EP is that evidence for associations can often be explained by overlap in superficial test characteristics. These superficial similarities could lead to the recruitment of a host of processes that can inflate correlations, whether or not there truly is a domain-general EP ability. For example, an early and foundational study captured individual differences in EP for features of low and high complexity (Haberman et al., 2015). Errors in EP judgments for the mean of color stimuli were correlated with errors for EP judgments of mean orientation, and people who were good at judging the average identity of faces were also good at judging their mean expression. This study also highlighted a dissociation, in that judgments for low-level features did not relate to those for faces. This pattern of results was used to suggest two distinct EP abilities that would apply to high-level features and low-level features, respectively.

While this work was well powered, one important limitation is that all correlations were between pairs of tests that had either the stimulus (e.g., both faces) or summary statistic (e.g., both mean) in common. This applies to all evidence of associations in this literature. For instance, errors in mean and diversity judgments correlate, but both pertain to the size of circles (Cha et al., 2022). Likewise, errors in EP judgments about cars and about birds correlate, but both types of judgments were mean estimation (Chang & Gauthier, 2021). Such correlations do not provide the kind of evidence that could begin to support an ability relevant to the broad diversity of EP tasks[1] used in the literature. This could be because EP is generally studied in labs that mostly conduct psychophysical studies of a more experimental nature – studies where comparisons of tasks that differ only in one way is a prudent approach. We propose that to find evidence for a "general ensemble processor", we need to adopt a different strategy. Therefore, in this work we seek a remote association between different EP tasks that do not overlap in their superficial characteristics (stimuli, task format and summary statistic).

1.3. Overlap between Ensemble Perception and Object Recognition

An early claim in the EP literature is that estimation of statistical summaries for groups of items cannot be explained by single item processing, meant to index object recognition. This was originally proposed because after seeing sets of circles of different sizes for 500 ms, observers were poor at deciding

[1] We use 'task' to denote a relatively broad category of trials that share common features (e.g., face matching and face learning are two face processing tasks) and 'test' to denote a specific measurement tool that can be used to compare individuals (e.g., Alice had a better score than Bob on our test of mean judgments for line orientation).

if a test circle had been presented or not, but surprisingly good at judging if a test circle was smaller or larger than the mean (Ariely, 2001). In other studies, although items are presented too fast for participants to report them individually, they yield above-chance EP judgments (e.g, Oriet & Corbett, 2008). This kind of work has led to the proposal that EP does not depend on the recognition of individual items (Whitney & Yamanashi Leib, 2018). Others have pointed out that successful EP performance based on weak signals for individual items to do not necessarily have to reflect the computation of abstract ensemble statistics. Instead, such EP judgments could reflect the combined influences of each noisy item, including their respective similarity to the central tendency (Harrison et al., 2021; also see Nosofsky, 1988, for a similar account of prototype effects in categorization).

Here, we remain agnostic as to whether EP judgments actually require abstraction of summary statistics (a case for actual abstraction is surprisingly difficult to make when comparing alternative models, Jeong & Palmeri, 2023). We do not make claims about strategies that participants may or may not use to make ensemble judgments. Some may use OR-based strategies, some of the time, especially if they are less adept at using EP-based strategies. Therefore, in addition to seeking proof of concept that performance on two very different EP tasks can correlate, we also wanted to address whether such a correlation could be accounted for by an ability that is distinct from OR. Some studies have tried to isolate EP abilities controlling for errors on single item judgments (Cha et al., 2022; Chang & Gauthier, 2021). They found that performance on different EP tests were correlated even after controlling for performance with individual items. In one large multivariate study measuring domain-general OR as a latent variable with novel and familiar objects (each with two tasks and three categories), an EP factor was estimated based on six average identity tests for different complex objects (Sunday et al., 2021). The correlations between the latent factor for EP and factors for OR estimated with novel or familiar objects were .62 and .67, respectively. This suggests overlap between abilities supporting judgments about individual objects and ensemble of objects. However, with almost 60% of the non-error variance remaining, the abilities are also distinct. But even in this large multivariate study, all EP tasks were of the same format (mean estimation for arrays of four items). For this reason, we cannot interpret their shared variance as providing unequivocal support for a domain-general EP ability. Given we chose EP tasks that together are positioned wide apart in the space of all possible EP tasks, we did not want to choose an OR task that was a highly similar version of one of the EP tasks (as in Haberman et al., 2015 for instance). Instead, we opted for an OR task that is highly representative of measurements for this ability. Specifically, we used a memory test structured similar to memory tests with faces (Duchaine & Nakayama, 2006) and familiar objects (McGugin et al., 2012) but using novel objects (Sunday et al., 2021).

1.4.Our study

We seek to estimate the magnitude of a correlation between tasks that can be attributed to a general EP ability, distinct from OR. Most importantly, we used two distinct EP tests that differ in their format (estimation vs. comparison), in the stimuli composing the ensembles (Transformer robots vs. two-tone meaningless blobs), and in the summary statistic to be estimated (mean vs. diversity). The Diversity Comparison Test with blob stimuli and the Mean Estimation Test with Transformers are shown in Figure 1. Despite their differences, both EP tests require explicit EP judgments that we hypothesize should tap into a domain-general EP ability. In addition, the differences between EP tasks limits the use of the same idiosyncratic strategy in both tasks.

It is worth making a few points about how we chose these tasks. First, many studies have designed EP tasks with the intention to rule out other strategies (e.g., by limiting presentation time or preventing foveation) in efforts to demonstrate that EP is a distinct process and to study it in isolation (Wolfe et al., 2015). We take a different approach by acknowledging that if EP is as ubiquitous, efficient, and compulsory as has been suggested (Whitney & Yamanashi Leib, 2018), such designs are not necessary to measure the influence of EP ability across a variety of more natural conditions. For instance, while preventing foveation does not prevent the ability to extract summary statistics (Wolfe et al., 2015), most EP studies allow foveation. Likewise, while presenting many items together and briefly does not abolish EP abilities (e.g., Yang et al., 2018), many EP studies show items one at a time in succession

(Haberman et al., 2009). Because we seek evidence for a general EP ability, we adopt an inclusive stance which assumes that EP abilities may contribute to performance on a vast number of tasks, including outside the laboratory (e.g., Szafir et al., 2016). We chose tasks that explicitly instruct participants to use summary statistics and for which integrating over several items would lead to relatively better performance. We chose tasks already used in prior studies on EP (Cha et al., 2021; Chang & Gauthier, 2021; Sunday et al., 2021), which have included demonstrations that on average, participants integrate across objects in these tasks (see supplemental analysis in Chang et al., 2022) and distribute attention broadly over items (Cha et al., 2021). However, we are not concerned that some participants, some proportion of the time, may fail to integrate and attempt to perform these judgments on the basis of a single item, as this would reflect poorer EP ability. Given our interest in individual differences, measuring a broad range of performance is important – our tasks encourage and benefit from using strategies that have been associated with EP in the literature and while some people on some trials may fail to use such strategies, it is most important that no strategy using focused attention on single items could lead to higher performance than trying to integrate over several objects. Other than this, our tasks were chosen to differ in terms of stimuli, summary statistic and task format, without any claim that they are particularly representative in any other way. The space of tasks used to measure EP is very large (Whitney & Yamanashi Leib, 2018) and our tasks are similar to many others in the literature (e.g., the diversity task with blobs is similar to other diversity judgments, e.g., Phillips et al., 2018, the Transformer mean estimation is similar to in format to tasks with faces and simple shapes, e.g., Haberman et al., 2015). In the discussion, we will return to the limitations that accompany our choice of tasks, but what is most novel in this work is the attempt to estimate shared EP contribution in two very different EP tasks.

In addition, based on earlier results, we expect that performance on the Mean Estimation Test with Transformers should also rely on OR (Sunday et al., 2021), while the Diversity Comparison Test with blob stimuli is not expected to tap into OR processes relevant to subordinate-level discrimination. As we describe below, the blob stimuli were created from random noise and do not as a group a common configuration of parts. Unlike the familiar or novel objects used in tests that tap into OR (Richler et al., 2019; Sunday et al., 2021), blob stimuli could not be morphed together to produce a recognizable member of the category. They also do not have plausible canonical orientation and are like different samples of the same texture. To test our assumption that the Blobs test and Transformer tests would be more similar in their use of EP than OR, we included a third test that should tap into OR, but not EP. We use the Novel Object Memory Test (NOMT) with asymmetric Greebles for this purpose. This test requires decisions about triplets of objects after studying an array of six objects (see Figure 1). It shares common variance with many other OR tests that vary in stimuli and task formats (e.g., Sunday et al., 2021). Therefore, in terms of superficial format it uses arrays of stimuli, but it does not require an explicit EP judgment – integrating over different objects would not facilitate performance on this task.

In summary, we use tasks that are sensitive to individual differences, and that have already been validated to tap into EP, OR or both OR and EP. Our results are the first to support the claim that a common EP ability, distinct from OR, underlies performance in two very different EP tasks.

2. Method

2.1. *Participants*

Two hundred participants completed this study for monetary reward through the online participant recruitment platform Prolific.co. Informed consent was obtained for each participant before they started the study. We expected pairwise correlations between our tasks to be around .4 (Chang & Gauthier, 2021). Correlations of this magnitude converge towards the true value (± .10) above 180 participants (Schönbrodt & Perugini, 2013), so we aimed to collect data from 200 participants. Power analyses that we conducted (Richler et al., 2019) for more complex models than the ones specified here indicate that this sample size would provide adequate power and precision for our confirmatory factor analyses. The study was approved by (*removed for blinding purposes*) Institutional Review Board. We report results on 198 participants (99% of recruited sample; 90 males and 108 females; Mean age = 25.82; SD = 6.42). Exclusions are discussed in Section 3.

2.2. Procedure

In one session, participants completed three tests in the following order: the Diversity Comparison Test with blob stimuli (DC-Blobs), the Mean Estimation Test with Transformers (ME-Transformers), and the Novel Object Memory Test with Greebles (NOMT-Greebles) (see Figure 1 for sample trials). To avoid any confounding effects of individual differences with varied presentation order, the trials of each test were given to all participants in a fixed order. Each test took approximately 10 minutes to complete. Participants were allowed to take breaks between tests.

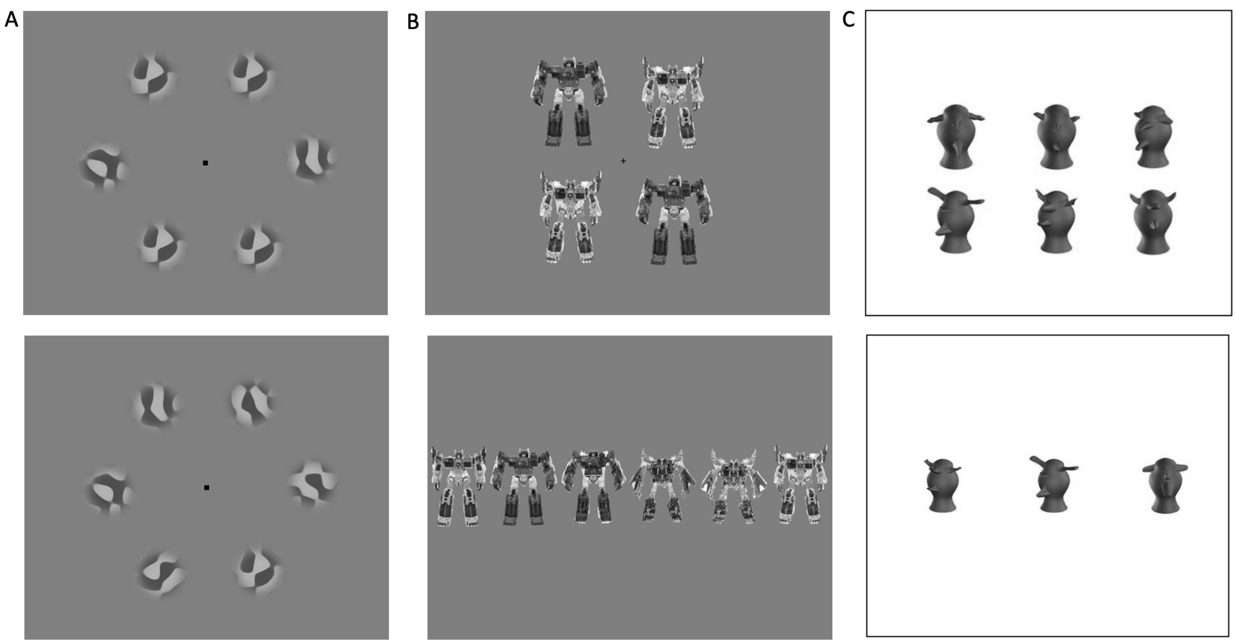

Figure 1. Example trials for the three tests in the study. In the Diversity Comparison Test with blob stimuli (A), participants judged which of two sequentially-presented arrays of blobs was the most diverse (the answer for this example is the second array). In the Mean Estimation Test with Transformers (B), participants saw an array of four Transformers and then clicked on the Transformer closest to the average of the group (the answer for this example is the first choice in the sequence). In the Novel Object Memory Test with Greebles (C) Participants studied six Greebles and then on each trial, chose which of three Greebles was one of the six Greebles that had been studied (the answer for this example is the middle Greeble – all Greeble were presented in the same green color).

2.2.3. Diversity Comparison Test with Random Blobs (DC-Blobs)

The Diversity Comparison Test with blobs required participants to estimate and compare the level of diversity in two arrays of random blobs (Cha et al., 2021). The test began with six practice trials (excluded from analysis), followed by two blocks of 64 experimental trials. The two formal blocks were separated with a 20-second mandatory break. Each trial began with a fixation dot presented for 800ms. Afterwards, two arrays of six blobs, placed equally distanced on an imaginary circle, showed up for 700ms sequentially with another fixation dot presented for 600ms in between. Participants were asked to select which of the two arrays had the larger number of different blobs by either clicking on a button specifying "First array was more diverse" or on a button specifying "Second array was more diverse."

A total of 150 images of blobs were used to create trials. These blob images were generated from 900 noise images processed with a band-pass filter centered at 3.6 cycles/image, each cropped with a circular window with blurred edges. Then, six subsets of 25 images each were selected while maximizing pixel-by-pixel similarity among images within the same subset and minimizing the similarity among images across different subsets (Cha et al., 2021). This resulted in 150 images that varied in pattern. No trial showed more than one image from the same subset, resulting in novel objects that were reasonably discriminable but limiting repetition across trials.

In each array, the six blobs could vary in their pattern. Thus, an array consisting of all distinctly patterned blobs would always be the more diverse array. The less diverse arrays could have three to four repeated blobs. Since this task is a two-alternative forced choice task, the chance level is .50. Accuracy was emphasized over response time with no time limit for making the judgment. Feedback on the accuracy was given for all trials. Due to the brief presentation of the stimuli, participants were encouraged to stay alert and fixate at the central dot so they could have a complete view of the arrays. Average accuracy was used to index the performance on this test. Cha et al. (2021) found that observers improved more over the course of the experiment on trials in which the repeated blobs were scattered as opposed to clustered (and thus easier to group). This indicated that the diversity comparison task is sensitive to the ability to distribute attention across all items when the repeated blobs were scattered. We only used the scattered condition for this study.

2.2.4. Mean Estimation Test with Transformers (ME-Transformers)

The Mean Estimation Test with Transformers measured how well participants perceived the average identity of images of multiple Transformers (Sunday et al., 2021) . During each trial, a study array of four Transformers arranged in 2 × 2 was presented with a central fixation for 1000 ms. The participants were instructed to fixate on the center cross so that they could view the four Transformers at once. After a 1000 ms inter-stimulus-interval, a response array of 6 Transformers showed up in a line and stayed on the screen until participants judged which Transformer most approximated to the mean identity of the four Transformers in the previously shown study array. Participants were instructed to respond as accurately as possible. Following four practice trials with feedback, a block of 50 experimental trials without feedback was performed. Note that one EP task had feedback, the other did not, which we kept from prior uses of these tasks because our goal was to have EP tasks differing in superficial characteristics.

All of the study arrays and response arrays were made of six base Transformers, derived from a circular morph continuum where pairs of three Transformer exemplars (i.e., Crankcase, Metroplex, and Fortress) were morphed. This morphing was performed by linearly interpolating key composite parts such as head, two arms, and legs to create a linear identity progression. To increase the variety of trials, and given an arbitrarily labeled circular morph series of items A,B,C,D,E and F, a study array either consisted of two instances of two Transformers (e.g., 2 Transformer As and 2 Transformer Cs, with the correct choice set to be transformer B) or two instances of one Transformer and its two flankers on the morph continuum (e.g., 2 Transformer Bs and 1 Transformer A and 1 Transformer C, with the correct choice set to be Transformer B). The ratio of these two kinds of trials was 18:32. Note that the correct answer may not have been the exact average identity of the group because there was no guarantee that all morphs were equidistant perceptually. The response arrays always consisted of the six base Transformers that constituted the study arrays lined up in identity progression. However, to prevent response bias in selecting a certain choice from the response arrays, the sequence of the six base Transformers was displayed in a random order from any Transformer on any given each trial (ABCDEF, BCDEFA, CDEFAB etc.). Chang & Gauthier (2021, supplemental section) compared participants performance on this task to an ideal observer's performance relying on repeated stimuli. Across four tasks, they found evidence that participants based their decision on integrating across more than repeated objects.

In line with Haberman et al. (Haberman et al., 2015), we calculated the mean absolute error in degrees. Each response was first scored as the absolute value of the difference between the response and the correct answer. For instance, if the correct answer on a trial was 5 and the participants responded 3, then the accuracy would be scored as a 2, for $|5 - 3| = 2$. Then, we multiplied this difference by 60 degrees to reflect that there were 6 exemplars expanding the total of 360 degrees in the morph space. An absolute difference of 0 degree represented the most accurate response, and a larger absolute difference indicated poorer performance. The absolute differences across all 50 experimental trials were then averaged to index the performance. For any given correct answer, the expected accuracy for a random choice was the sum of the distance between the correct answer and each option, i.e., 0 (the correct answer is chosen) + 1, + 2, + 3, + 2, + 1, divided by 6 possible options, multiplied by 60 degrees. Accordingly, chance performance for this test is 90 degrees. For the confirmatory factor analysis (CFA), we used the original scores before transformed into degrees, so that scaling was more similar across all test scores.

2.2.4. Novel Object Memory Test with Greebles (NOMT-Greebles)

The Novel Object Memory Test with Greebles is a learning-based object recognition test that requires participants to first study 6 target Greebles and then recognize them among distractor Greebles (Richler et al., 2019; Sunday et al., 2021). The test entailed three study-test blocks that each started with a 20-second study phase, followed by 6, 18, and 24 untimed test trials, respectively. During the study phase, participants learned 6 target Greebles that were presented simultaneously in a 2×3 arrangement. During the test phase, a target Greeble appeared alongside two distractor Greebles, and the participants were asked to click on the target Greeble. In the first test phase, there were 6 same-viewpoint trials where images of the target Greebles were identical to the ones shown in the study. In the second block's test phase, participants completed 12 more same-viewpoint trials and another 6 same-viewpoint trials in which Gaussian noise was added to all three Greebles. In the third block's test phase, participants were instructed that the target Greebles would be presented in a different viewpoint from what they had studied during the remaining trials. Following the last 20-second study phase, participants completed 12 such different-viewpoint trials and 12 more different-viewpoint trials with added Gaussian noise. Two additional trials were inserted in the second and third block to ensure attention to instructions. In these attention check trials, three place holders were presented with one marked "Click here" and the other marked "Do not click here". The test was not timed and participants were encouraged to be as accurate as possible. No feedback was given for the entire test. Performance on the test was indexed by percent correct over the total of 48 trials. As a 3 Alternative-Forced-Choice task, this test had a chance level of .33. Although there were three Greebles presented at the same time on each test trial, similar to the arrays used in EP tests, the NOMT required the recognition of individual objects and did not require any EP judgment. Therefore, participants might deploy their attention differently than in the other two tasks, studying each Greeble one at a time.

3. Analyses

3.1 Exclusions

Prior to data analyses, we screened all participants based on their accuracy on the two attention checks. Among 200 participants, one participant missed both attention checks and performed at chance or below chance on all three tasks (mean accuracy ≤ .50 for DC-Blobs, mean absolute error ≥ 90° for ME-Transformers, and mean accuracy ≤ .33 for NOMT) and thus was excluded. Five other participants missed one attention check, but only one of them performed below chance on more than one task. Therefore, all data from only two participants were excluded from further analyses for failing at least one attention check and not performing above chance on at least two tasks. The results reported here are derived from the remaining 198 participants (99% of recruited sample; 90 males and 108 females; Mean age = 25.82; SD = 6.42).

3.2 Reliability and zero-order correlations

We report the reliability (internal consistency) of our measurements using both Cronbach's α and Guttman's λ_2. Guttman's λ_2 is more robust than Cronbach's α when measures are comprised of multiple factors and relative to other indices of internal consistency, it offers a compromise between precision and bias (Oosterwijk et al., 2016). We report the zero-order correlations between the three tests. We used JASP statistics software (JASP Team, 2022) for Bayesian analyses. The log Bayes Factor (BF_{+0}) was used to test the possibility in favor of the presence of a correlation (relative to no correlation). We expected small to moderate positive correlations, therefore we used directional hypotheses (H+: positive correlation vs H0: point null). Reflecting these expectations for a small to moderate positive correlation, we used a prior described by a beta-distribution centered around zero and using the JASP parameter of .3, but using only the positive side of the distribution. This corresponds to a prior probability of 80% that the correlation coefficient lies between 0 and .5. For all correlations, we assessed robustness to priors and found them to be robust for all values above the minimum parameter allowable in the software (.003).

We interpret $\log BF_{+0}$ with values of .5 to 1 as substantial evidence for H0, values of 1 to 2 considered strong evidence for H0 and values above 2 are considered decisive evidence for H0. To index uncertainty of our point estimates, we used the 95% highest posterior densities as credible intervals (95% CI).

3.3 Analysis of error for the Transformer task

The Transformer task is the most complex because we expect it to have a contribution from both OR and EP. A reviewer was concerned that participants may perform the task based only on one of the presented objects. In this task, there were six response options (A,B,C,D,E,F) showing objects separated by 60 degrees in a circular morph space (the options were always shown in order, starting from a randomly selected option). Assuming that the correct answer is the second response option, the six responses therefore correspond to errors of 60°, 0°, 60°, 120°, 180° and 120°, respectively.

There were 32 trials in which the correct answer was presented twice in the array (A,B,B,C trials, where B is the answer). The distribution included the following proportions for answers that were 0°, 60°, 120°, or 180° away respectively from the correct answer: 40.2%, 18.4%, 8%, and 7.1% (the response proportion for 60° and 120° is divided in half because of the symmetrical response array). For the best performers (those with a median error ≤ 58.8°), the distribution was even more peaked (52.4%, 18.3%, 3.9% and 3.2%).

There were also 18 trials in which the correct answer as not presented in the array (A,A,C,C trials, where B is the answer). These trials are particularly helpful for us to determine whether participants select only objects presented in the array, or a single object, without integration. This is because neither the correct option (B) nor those that were 120° away from correct (D or F) were presented in the array, and both of these options are 60° away from the objects presented in the array. The response distribution was as follows for answers that were 0°, 60°, 120°, or 180° away respectively from the correct answer: 20.1%, 27.2%, 9.6%, and 6.2%. For the best performers, the distribution was 52.4%, 18.3%, 3.9% and 3.2%. Accordingly, participants were at least twice as likely to select B as they were to select D or F (and 13 times as likely for the best performers), suggesting that they did not answer on the basis of a single sampled object or even simply attending to a pair of identical objects.

3.4 Confirmatory Factor Analysis

We used confirmatory factor analysis (CFA; Tomarken & Waller, 2005), a special case of structural equation modeling (SEM), to specify latent variable models to test our hypothesis. Our prediction is that performance on the tasks that depend on the same ability (either EP or OR) will be more strongly correlated than those tasks that depend on different abilities. SEM is a powerful approach to compare different models of the data. Most importantly, we compare a model in which all pairwise correlations are the same to a model in which the critical correlation is lower than the other two. A regression approach or simply comparing correlations would test these hypotheses using average scores as a measure of performance in each task. Such approaches assume variables are measured without error, and thus in practice measurement error is confounded with performance and can affect the results (e.g., a correlation can be lower than another one because of more error variance). In contrast, with CFA we can model measurement error to estimate the relations among the tasks in a manner that is relatively free from its influence, and also brings the ability to specify and test the absolute and relative fit of competing models. Typically, latent variables in CFA are estimated using several tasks or indicators. But latent variables can be created by averaging groups of items (here trials) into parcels that function as indicators of a construct. This approach provides several advantages over total scores (limiting influence of measurement error and providing more valid estimates of relations between constructs, (Little et al., 2013). How the parcels are constructed is less important than the fact that they are used and when sample size is higher than 150 and factor loadings are >.6 (conditions met here), different ways of parceling items converge on the same centroid for the constructs (Little et al., 2013; Sterba & MacCallum, 2010). Readers not accustomed to SEM may consider the role of parcels as similar to analyzing blocks of trials instead of individual trials when looking at performance over time, to reduce noise, or dividing trials into groups to compute split-half reliability.

Refer to Figure 3 for an illustration of our model structure for CFA, a correlated 3-factor model. Among the three tests in this study, DC-Blobs and ME-Transformers are relevant to EP judgments whereas ME-Transformers and NOMT are relevant to OR. In our model, each test was represented as a factor that is indicated by multiple parceled average scores of that particular test. Specifically, we divided each test into parcels of trials to create multiple indicators for each factor. We sought to create 4 to 5 parcels per task, homogeneously when all trials were similar, or using blocks when they existed. For the DC-Blobs factor, we created 4 parcels by dividing the 128 trials evenly by 4. Therefore, each parcel had 32 trials (e.g., parcel 1 included the 1st, 5th, 9th, …, and 125th trials, parcel 2 the 2nd, 6th, 10th, ..., and 126th trials) and for each parcel, we used the average percent accuracy score across these 32 trials to serve as an indicator for the factor DC-Blobs. Using the same method, the 50 ME-Transformers trials were partitioned into 5 parcels, each including 10 trials (e.g., parcel 1 included the 1st, 6th, 11th, …, and 46th trials) and summarized as the mean absolute error for each parcel. These 5 indicators loaded on the factor for ME-Transformers. To create the indicators for the factor NOMT, we divided the total 48 trials into 4 parcels based on different testing conditions. Specifically, one parcel included 12 trials in which the test Greebles were the same image as those shown during study; one parcel included 6 trials in which test Greebles in the original viewpoint were presented with added visual noise; one parcel included 12 trials in which test Greebles were shown in a different viewpoint from the originally presented in study phase; and one parcel included the last 12 trials where test Greebles were shown in a different viewpoint and with visual noise. The mean percent accuracies of these 4 parcels were used as 4 indicators that load onto the factor NOMT.

Since our main goal was to assess their relationship, the three factors were allowed to correlate. To answer the primary question regarding whether the data reflects separate EP and OR abilities, we specified three alternative CFA models. In the first model (Model 1), the parcels for the three tests loaded onto their corresponding task factors (e.g., the 4 parcels of DC-Blobs loaded onto the DC-Blobs factor). Factor variances were standardized, all factor loadings were estimated freely and three covariances between the factors (identical to correlations given the standardization) were also estimated freely. Model 1 served as a base model to examine how the correlated 3-factor model structure fits the actual data. We expected the three factors to correlate significantly based on our correlation analyses (Section 4.2) and previous reports (Cha et al., 2022, Chang & Gauthier, 2021a, Sunday et al., 2022) that showed overlap between EP and OR.

Next, our main interest was testing two alternative models, Models 2 and 3, both of which were built on Model 1 but posited additional constraints on the correlations between the three measure factors. That is, we predicted correlations between EP factors (i.e., *r*(DC-Blobs, ME-Transformers)) and between OR factors (*r*(ME-Transformers, NOMT) to be larger than the correlation between EP and OR factor (*r*(DC-Blobs, NOMT). In Model 2, factors that we assumed to correlate more strongly were constrained to be equal, i.e., *r*(DC-Blobs, ME-Transformers) = *r*(ME-Transformers, NOMT). In Model 3, all three correlations between factors were constrained to be equal, i.e., *r*(DC-Blobs, ME-Transformers) = *r*(ME-Transformers, NOMT) = *r*(DC-Blobs, NOMT). Model 3 is nested in Model 2 since it is almost identical to Model 2 except that one parameter, the correlation between *r*(DC-Blobs, NOMT), was further restricted to be equal to the other factor correlations. By comparing Models 2 and 3, we can directly test whether all three factors are equally correlated or not. To clarify, we are not particularly interested in the equality of any two correlations, so this comparison of models allows us to test whether some correlations are stronger than the others. Our hypothesis predicts that correlations between factors influenced by EP and between factors influenced by OR should be larger than, and cannot be constrained to be equal to, the correlation between tasks that do not share either of these influences. This would be supported by a comparative fit between Model 1 and Model 2 and a better fit of Model 2 over Model 3.

All CFA models were tested using the R library lavaan (Rosseel, 2012) with maximum likelihood estimation used to estimate model parameters. We used the MLR estimator in lavaan, which provides robust standard errors and a scaled chi-square statistic that corrects for violations of multivariate normality, an assumption associated with the standard maximum likelihood estimator. We used several goodness of fit indices to evaluate models. The chi-square test of exact fit is the standard measure of model fit that MLR estimator generates. A nonsignificant result for the scaled χ^2 statistic indicates a good model fit but it is highly sensitive to sample size (i.e., rejects even well-fitting models with large sample size) and provides only a binary decision (accept-reject). Consistent with Richler et al. (2019) and Sunday et al. (2021), we also used other indices to characterize model fit, including the Comparative Fit Index (CFI) and the root mean-squared error of approximation (RMSEA). The CFI reflects the incremental or comparative fit of the target model relative to a baseline model of independence in which all the covariances among the observed indicators are fixed at 0. CFI values vary from 0 to 1, with values closer to 1 indicating better fit. We used a CFI larger than .95 to infer a good model fit. The RMSEA reflects the degree to which a model fits approximately in the population, we focused on the values of .05 and .08 to differentiate good fit and mediocre fit, respectively.

To compare the relative fit of different models, we use the Akaike Information Criterion (AIC; Akaike, 1973) and the Bayesian Information Criterion (Raftery, 1995). These indices penalize for model complexity (number of free parameters in a model). The smaller these indices, the better a model is in terms of goodness of fit and model parsimony. To compare nested models (i.e., between M1 and M2, or between M2 and M3), we ran likelihood ratio tests (LRT in lavaan) with scaled chi-square difference (Satorra & Bentler, 2001). A non-significant result suggests that the restriction of parameter(s) in the nested model does not impair model fit.

4. Results

4.1. Descriptive statistics, reliability and normality

Table 1 summarizes the descriptive statistics and reliability estimates of the three tests. Each test demonstrated a good range of performance and good reliability.

Table 1. Descriptive statistics, reliability and normality indices

	Mean (SD)	**[Min, Max]**	**Skewness**	**Kurtosis**	**Reliability (α, λ_2)**
DC-Blobs	.70 (.09)	[.33, .95]	-.48**	2.43*	(.92, .92)
ME-Transformers	-58.58° (13.67°)	[-99.60°, -26.40°]	-.22 (p = .2)	2.18***	(.82, .83)
NOMT-Greebles	.55 (.12)	[.08, .90]	.03 (p = .85)	2.65 (p = .31)	(.81, .82)

Note. Results from DC-Blobs are percent accuracy (chance performance = .50); ME-Transformers results are mean absolute error in degrees multiplied by -1 (chance performance = -90°), to keep all correlations positive; and NOMT-Greebles results are percent accuracy (chance level = .33). Positive value for skewness indicates positively skewed distribution; a value larger than 3 for kurtosis indicates leptokurtic distribution, a value smaller than 3 indicates platykurtic distribution, compared to normal distribution (skewness = 0, kurtosis = 3). Results from D'Agostino skewness test and Anscombe-Glynn kurtosis test are reported, * $p < .05$, ** $p < .01$, *** $p < .001$.

4.2. Correlations

Results showed decisive evidence for a positive correlation between all pairs of tests (Table 2). Any general source of variation (e.g., fluid intelligence, motivation, arousal) could account for some of the variance common to all three tests. Note however that this would be true for any pair of tasks, including pairs of EP tasks that are highly similar in format, stimuli and/or summary statistics. That is, there is no reason why any general influence would play a bigger role in this correlation than in prior cases where superficial task properties were also likely to inflate the correlation. Therefore, it is interesting that our two very different EP tasks show a correlation of a magnitude similar to that found for pairs of tasks with different stimuli but the same format (e.g., Haberman et al., 2015, $r = .54$ for mean estimation of the color of four circles or the orientation of four Gabor patches; Chang et al., 2022, $r = .63$ between mean estimation for four birds or four planes), or different summary statistics for the same stimuli (Cha et al., 2022, $r = .45$ between estimating of the mean size vs. variance of circles).

Such general sources of variation would also contribute to the NOMT-Greebles performance, and this is consistent with the finding that the OR task correlates with both EP tasks. However, we hypothesized that the DC-Blobs and ME-Transformers tasks would share an influence from an EP ability that would not influence NOMT-Greebles performance. Partial support for this claim comes from the fact that a positive correlation between DC-Blobs and ME-Transformers remained decisive (partial $r = .42$, 95% C.I. = [.30, .53], log $BF_{+0} = 17.74$; see Figure 2), when controlling for NOMT-Greebles. However, as mentioned in the Analyses section, correlation and regression assume that variables are error free, which can bias and underestimate the relationship between them (Bollen, 2002). A better option is to parcel the results of each task to measure relations between latent variables instead, as used in the next section, to apply CFA to test our hypothesis.

Table 2. Pairwise Pearson correlations between the three tests, with log BF_{+0}

		DC-Blobs	ME-Transformers
ME-Transformers	Pearson's r	.52	-
	logBF_{+0} (95% CI)	28.58 (.41, .62)	-
NOMT-Greebles	Pearson's r	.37	.47
	logBF_{+0} (95% CI)	12.17 (.25, .49)	21.93 (.35, .57)

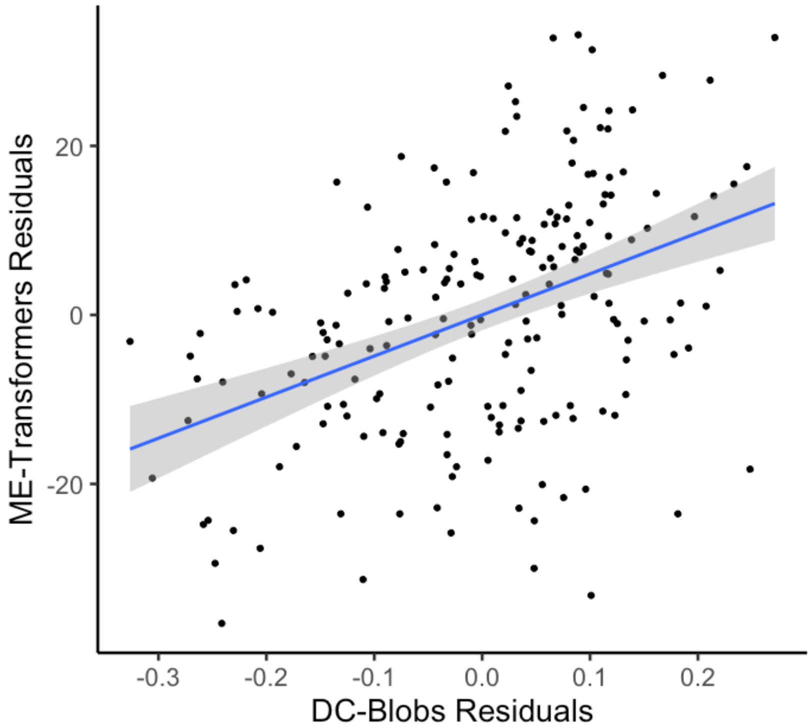

Figure 2. Scatterplot of DC-Blobs residuals and ME-Transformers residuals.

4.3. Confirmatory Factor Analysis

Table 3 summarizes the model fit indices for the three models we compared. Model 1, a simple correlated 3-factor model, showed evidence of good fit (CFI = .991 and RMSEA = .029). Because the factors were standardized, the covariances between the factors equal their correlations. All three pairwise correlations were positive and significant. Speaking more directly to our question, the correlation between factors that we expected to contribute to EP (r(DC-Blobs, ME-Transformers) = .59, $p < .0001$) and OR (r(ME-Transformers, NOMT) = .56, $p < .0001$) were numerically larger than the correlation between factors that tap into different abilities (r(DC-Blobs, NOMT) = .41, $p < .0001$). But can we consider this difference to be significant?

Table 3. Summary of model fit indices for the structural equation models tested in this study.

Model	Description	df	Robust $\chi 2$	p	Robust CFI	Robust RMSEA (90% CI)	AIC	BIC
1	Freely correlated 3-factor model	62	72.27	.18	.991	.029 (0, .053)	-1511.2	-1415.8
2	Correlated 3-factor model with equality constraint: *r*(EP factors) = *r*(O factors)	63	72.40	.20	.992	.028 (0, .054)	-1513.0	-1420.9
3	Correlated 3-factor model with equality constraint: *r*(EP factors) = *r*(O factors) = *r*(DC-Blobs, NOMT)	64	78.37	.11	.988	.034 (0, .058)	-1508.8	-1420.0

Note. EP factors are DC-Blobs and ME-Transformers; O factors are ME-Transformers and NOMT; CFI = Comparative Fit Index (good fit indicated as > .90); RMSEA = root mean square error of approximation (the cutoffs for excellent, good, and mediocre fit are .01, .05, and .08); AIC = Akaike Information Criteria (better fit indicated by smaller AIC); BIC = Bayesian Information Criteria (better fit indicated by smaller BIC).

To answer this question, we compared Model 2 and Model 3. This allows us to test if the abilities that we hypothesized to be associated with EP and OR were best modeled with equal pairwise correlations between the factors (Model 3) or stronger correlations for the pairs that tapped into a common postulated ability (Model 2). Recall that Model 2 is built on Model 1 but with the added constraint that correlations of the two EP-influenced factors (DC-Blobs and ME-Transformers) and the two OR-influenced factors (ME-Transformers and NOMT) are equal. Both fit indices suggest good fit of Model 2 (CFI = .992 and RMSEA = .028, Table 3). Furthermore, adding the equality constraints on the correlation between EP-influenced factors and the correlation between OR-influenced factors did not impair model fit (Model 1 vs. Model 2 in Table 4). The constrained correlations between EP-influenced factors and OR-influenced factors were estimated to be .58 ($p < .0001$) and the correlation between DC-Blobs and NOMT was .41 ($p < .0001$).

In Model 3, we further constrained all three pairwise correlations between factors to be the same (estimated to be .52, $p < .0001$). Model 3 also showed evidence of good fit (CFI = .988 and RMSEA = .034, Table 3). However, the additional 3-way correlation constraint impaired model fit, as indicated by a significant chi-squared difference test (Model 1 vs. Model 3 in Table 4). Critically, the chi-squared difference test contrasting between Model 2 and Model 3 allows us to reject Model 3. This indicates that the correlation between DC-Blobs and NOMT is smaller than the correlations between the other two pairs of tests. Model 2 is illustrated in Figure 3.

Table 4. Summary of the model comparison results.

Models	*df*	*χ2 difference*	*p*	*Decision*
1 vs. 2	1	.20	.66	The equality constraint does not impair model fit *r*(EP factors) and *r*(O factors) can be constrained equal
1 vs. 3	2	6.00	.049*	Model fit was impaired if *r*(EP factors), *r*(OR factors) and *r*(*DC-Blobs*, *NOMT*) was constrained to be equal
2 vs. 3	1	6.06	.01**	*r*(*DC-Blobs*, *NOMT*) cannot be constrained equal to *r*(EP factors) and *r*(OR factors)

Note. EP factors are *DC-Blobs* and *ME-Transformers*; O factors are *ME-Transformers* and *NOMT*

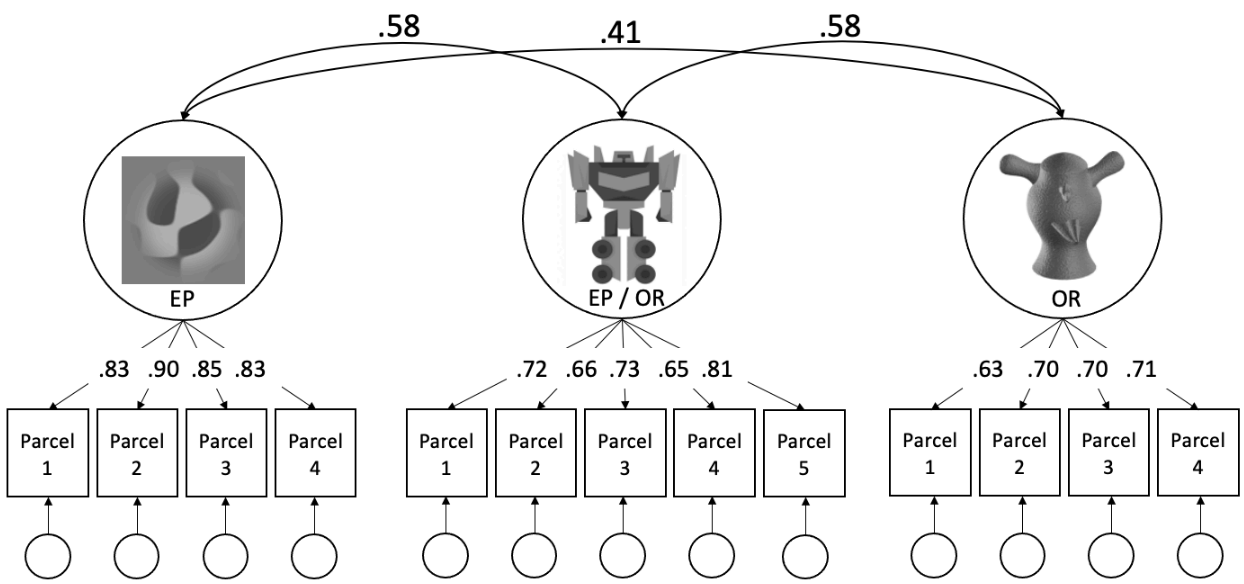

Figure 3. Graphical Representation of Model 2 (Standardized Solutions). Three factors are represented by large circles while observed scores are represented by rectangles and are linked to the corresponding factors with direct arrows. The numbers associated with these direct arrows are factor loadings (ranging from 0 to 1) that show how well the indicators measure the relevant factor. Measurement errors associated with the indicators are illustrated as small circles. The correlations (rather than covariances, because factor variances are standardized) between factors are shown as double-headed arrows that link pairs of the three factors.

5. General Discussion

We used individual difference methods to examine the abilities underlying different EP judgments and OR. In a large sample, we found decisive support for a positive correlation between performance on two very different EP tests, even after controlling for a third test that also presented groups of objects at the same time (six Greebles at study and three Greebles on each test trial) but required OR and not EP. CFA results found a larger correlation between a pair of tests tapping into EP or a pair of tasks both tapping OR, than tests that did not share these influences, supporting our hypotheses about two different abilities contributing to these tasks. Evidence for different abilities is not the same as evidence for different processes, but it can be a powerful guide to investigate underlying mechanisms (Wilmer, 2008).

By design, our two EP tests were very different: one measured mean estimation for ensembles of complex objects and the other measured diversity comparison for ensembles of much simpler stimuli. Prior work supported a common ability for judging mean and variance on the same objects (e.g., Cha et al., 2022) or for judging different features of the same objects (e.g., Kacin et al., 2021). We expand beyond this to offer evidence for a common ability even when no superficial task or stimulus constraint links the two tests. Such an ability has been implied by using the term ensemble perception for a large number of phenomena (Whitney & Yamanashi Leib, 2018). However, we argue that to date, the construct validity of such a broad ability has not been tested. When Haberman et al. (2015) proposed that EP was at least divided into mechanisms for high and low-level stimuli, they left many possibilities open. One possibility is that while the mechanisms are entirely domain-specific, stimuli that reside closer in representational space share a larger degree of perceptual noise. To the extent that many EP tasks using stimuli of a wide range of complexity converge on an ability that is distinct from OR, this could be more difficult to explain on the basis of noise widely distributed in the visual system. Sooner or later, most subfields of cognitive psychology must address assumptions about individual differences explicitly, and many of these assumptions about the phenomena we study do not pan out. For instance, different measures of holistic processing (Gauthier, 2020) or inhibition (Rouder & Haaf, 2019) fail to converge.

While only a starting point, our findings provide important insights. Evidence for a common EP ability for blob and transformer ensembles suggests it may be premature to assume that EP abilities between complex objects and simpler features do not overlap and to only focus on exploring mechanisms within each of these domains. Exploring this question was not the goal of the present work. Blobs are certainly more complex than the simple features in Haberman et al. (2015), but they do not form an object category with a fixed number of parts in a common configuration, like faces, cars or Transformers. In that sense they are more akin to small portions of the same texture. It also seems plausible that EP judgments about blobs could overlap with those for much simpler features. Studies that include stimuli like blobs, that share properties with both complex objects and simple features, may facilitate a more complete understanding of EP abilities.

Consistent with Chang & Gauthier (2021) and Sunday et al. (2021), our results suggest that EP judgments correlate with OR but also rely partly on a distinct ability. However, one caveat of our design is that both OR tasks used complex categories (Transformers and Greebles) while the two EP tasks spanned from simpler (blobs) to more complex (Transformers) categories. We did not predict that the correlation between the two OR tasks should be higher than that between the two EP tasks, because we did not expect the stimulus category to be very important in driving the EP ability. In prior work comparing EP judgments for blobs to EP judgments with faces and cars (Cha et al., 2021), we found very similar results for all three categories, with no effect of using a large or a small set of stimuli. The only difference between categories was evidenced by improvements during the experiment with blobs, but not with the more familiar categories. Since the complex objects used here were either novel (Greebles) or of limited familiarity relative to cars and faces, familiarity was not expected to have a strong influence.

As we provide the first evidence that very different tasks in the large space of EP phenomena can tap a common ability, this work should be most useful as a motivation to adopt new methods to investigate this question. We have argued that any pair of tasks that are similar along specific dimensions (e.g., stimuli or task format) could correlate on that basis alone, without reflecting a general ability. And while a correlation between a pair of tasks that are distant in a multidimensional task space may provide such evidence, it offers no information as to the structure of abilities within this space. Putting it another way, the observed correlation between two distant tasks may not be representative of the correlation that would be obtained between other pairs of tasks that vary in other ways. Accordingly, we can provide some advice for future work that would seek to investigate EP abilities. First, single tasks cannot be used to represent general constructs. For example, inferences about EP for complex features should not be based only on face tasks (Haberman et al., 2015). In the present work, we used latent variables as a tool to control for measurement error for a single task, but latent variables that represent the shared covariance between different tasks thought to tap a similar ability are a powerful way to measure abilities. Second, we used model comparison in CFA to test a hypothesis about pairwise correlations but this approach is a powerful means of testing more complex models about the structure of abilities. Putting these two recommendations together, we believe that a robust understanding of individual differences in EP will require the adoption of large scale multivariate studies which have been useful to test hypotheses about cognitive (e.g., Engle et al., 1999) and perceptual (Richler et al., 2019) constructs. The ultimate goal is to grow our theoretical understanding by eventually bridging experimental and individual differences studies. In these efforts, it remains important for tests of experimental influences and measurement of individual differences to apply the best methods available in each case, because conventions in one tradition can be counterproductive in the other (Hedge et al., 2018).

6. Conflict of Interest statement.

On behalf of all authors, the corresponding author states that there is no conflict of interest.

7. Compliance with Ethical Standards

The study was approved by (*removed for blinding purposes*) Institutional Review Board, and it was completed in accordance with the ethical standards of the 1964 Helsinki Declaration and later amendments. Funding sources provided on title page for blinding purposes.

8. Data availability statement.

Provided on title page for blinding purposes.

9. References

Akaike, H. (1973). Information theory and an extension of the maximum likelihood principle. *Proc. 2nd International Symposium on Information Theory, 1973*, 267–281.

Ariely, D. (2001). Seeing sets: Representation by statistical properties. *Psychological Science*, *12*(2), 157–162.

Balas, B., Nakano, L., & Rosenholtz, R. (2009). A summary-statistic representation in peripheral vision explains visual crowding. *Journal of Vision*, *9*(12), 13–13.

Bollen, K. A. (2002). Latent variables in psychology and the social sciences. *Annual Review of Psychology*, *53*(1), 605–634.

Brady, T. F., & Alvarez, G. A. (2015). Contextual effects in visual working memory reveal hierarchically structured memory representations. *Journal of Vision*, *15*(15), 6–6.

Brennan, P. C., Gandomkar, Z., Ekpo, E. U., Tapia, K., Trieu, P. D., Lewis, S. J., Wolfe, J. M., & Evans, K. K. (2018). Radiologists can detect the 'gist'of breast cancer before any overt signs of cancer appear. *Scientific Reports*, *8*(1), 1–12.

Cha, O., Blake, R., & Gauthier, I. (2021). The role of category- and exemplar-specific experience in ensemble processing of objects. *Attention, Perception, & Psychophysics*, *83*(3), 1080–1093. https://doi.org/10.3758/s13414-020-02162-4

Cha, O., Blake, R., & Gauthier, I. (2022). Contribution of a common ability in average and variability judgments. *Psychonomic Bulletin & Review*, *29*(1), 108–115.

Chang, T.-Y., & Gauthier, I. (2021). Domain-general ability underlies complex object ensemble processing. *J. Exp. Psychol. Gen.*, *Advance online publication*. https://doi.org/10.1037/xge0001110

Duchaine, B., & Nakayama, K. (2006). The Cambridge Face Memory Test: Results for neurologically intact individuals and an investigation of its validity using inverted face stimuli and prosopagnosic participants. *Neuropsychologia*, *44*(4), 576–585. https://doi.org/10.1016/j.neuropsychologia.2005.07.001

Engle, R. W., Tuholski, S. W., Laughlin, J. E., & Conway, A. R. A. (1999). Working memory, short-term memory, and general fluid intelligence: A latent-variable approach. *Journal of Experimental Psychology: General*, *128*(3), 309–331. https://doi.org/10.1037/0096-3445.128.3.309

Gauthier, I. (2020). What We Could Learn About Holistic Face Processing Only From Nonface Objects. *Curr. Dir. Psychol. Sci.*, *29*(4), 419–425.

Haberman, J., Brady, T. F., & Alvarez, G. A. (2015). Individual differences in ensemble perception reveal multiple, independent levels of ensemble representation. *J. Exp. Psychol. Gen.*, *144*(2), 432–446. https://doi.org/10.1037/xge0000053

Haberman, J., Harp, T., & Whitney, D. (2009). Averaging facial expression over time. *J. Vis.*, *9*(11), 1–13. https://doi.org/10.1167/9.11.1

Hansmann-Roth, S., Kristjansson, A., Whitney, D., & Chetverikov, A. (2021). Limits of perception and richness of behaviour: Dissociating implicit and explicit ensemble representations. Scientific Reports, 11, 3899.

Harrison, W. J., McMaster, J. M. V., & Bays, P. M. (2021). Limited memory for ensemble statistics in visual change detection. *Cognition*, *214*, 104763. https://doi.org/10.1016/j.cognition.2021.104763

Hedge, C., Powell, G., & Sumner, P. (2018). The reliability paradox: Why robust cognitive tasks do not produce reliable individual differences. *Behav. Res. Methods*, *50*(3), 1166–1186.

JASP Team. (2022). *JASP (0.16.3)*. https://jasp-stat.org/

Jeong, J., & Palmeri, T. (2023). *A computational modeling framework for ensemble perception*. Vision ScienceS Society.

Kacin, M., Cha, O., & Gauthier, I. (2021). Ensemble coding of average length and average orientation are correlated. *Vision Research*, *187*, 94–101.

Khvostov, V. A., & Utochkin, I. S. (2019). Independent and parallel visual processing of ensemble statistics: Evidence from dual tasks. *Journal of Vision*, *19*(9), 3–3.

Kline, R. B. (1998). Structural equation modeling. *New York: Guilford*.

Little, T. D., Rhemtulla, M., Gibson, K., & Schoemann, A. M. (2013). Why the items versus parcels controversy needn't be one. *Psychological Methods*, *18*(3), 285.

Logan, G. D. (1988). Toward an instance theory of automatization. *Psychol. Rev.*, *95*(4), 492–527.

McGugin, R. W., Richler, J. J., Herzmann, G., Speegle, M., & Gauthier, I. (2012). The Vanderbilt Expertise Test reveals domain-general and domain-specific sex effects in object recognition. *Vis. Res.*, *69*, 10–22. https://doi.org/10.1016/j.visres.2012.07.014

Nosofsky, R. M. (1988). Exemplar-based accounts of relations between classification, recognition, and typicality. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *14*(4), 700.

Oosterwijk, P. R., Ark, L., & Sijtsma, K. (2016). Overestimation of reliability by Guttman's λ 4, λ 5, and λ 6 and the greatest lower bound. *The Annual Meeting of the Psychometric Society*, 159–172.

Oriet, C., & Corbett, J. (2008). Evidence for rapid extraction of average size in RSVP displays of circles. *Journal of Vision*, *8*(6), 13–13.

Phillips, L. T., Slepian, M. L., & Hughes, B. L. (2018). Perceiving groups: The people perception of diversity and hierarchy. *Journal of Personality and Social Psychology*, *114*(5), 766.

Raftery, A. E. (1995). Bayesian model selection in social research. *Sociological Methodology*, 111–163.

Richler, J. J., Tomarken, A. J., Sunday, M. A., Vickery, T. J., Ryan, K. F., Floyd, R. J., Sheinberg, D., Wong, A. C. N., & Gauthier, I. (2019). Individual differences in object recognition. *Psychological Review*, *126*(2), 226–251. https://doi.org/10.1037/rev0000129

Robinson, M. M., & Brady, T. (2022). *A quantitative model of ensemble perception as summed activation in feature space*. PsyArXiv. https://doi.org/10.31234/osf.io/k3d26

Rouder, J. N., & Haaf, J. M. (2019). A psychometrics of individual differences in experimental tasks. *Psychonomic Bulletin & Review*, *26*(2), 452–467.

Satorra, A., & Bentler, P. M. (2001). A scaled difference chi-square test statistic for moment structure analysis. *Psychometrika*, *66*(4), 507–514. https://doi.org/10.1007/BF02296192

Schönbrodt, F. D., & Perugini, M. (2013). At what sample size do correlations stabilize? *Journal of Research in Personality*, *47*(5), 609–612. https://doi.org/10.1016/j.jrp.2013.05.009

Schubert, A.-L., Löffler, C., & Hagemann, D. (2022). A neurocognitive psychometrics account of individual differences in attentional control. *Journal of Experimental Psychology: General*, *151*(9), 2060–2082.

Sterba, S. K., & MacCallum, R. C. (2010). Variability in parameter estimates and model fit across repeated allocations of items to parcels. *Multivariate Behavioral Research*, *45*(2), 322–358.

Sunday, M. A., Tomarken, A. J., Cho, S.-J., & Gauthier, I. (2021). Novel and familiar object recognition rely on the same ability. *J. Exp. Psychol. Gen.*, *13*, 93–110.

Szafir, D. A., Haroz, S., Gleicher, M., & Franconeri, S. (2016). Four types of ensemble coding in data visualizations. *Journal of Vision*, *16*(5), 11. https://doi.org/10.1167/16.5.11

Tomarken, A. J., & Waller, N. G. (2005). Structural Equation Modeling: Strengths, Limitations, and Misconceptions. *Annual Review of Clinical Psychology*, *1*(1), 31–65. https://doi.org/10.1146/annurev.clinpsy.1.102803.144239

Utochkin, I. S., Choi, J., & Chong, S. C. (2023). A population response model of ensemble perception. *Psychological Review*.

Whitney, D., & Yamanashi Leib, A. (2018). Ensemble Perception. *Annu. Rev. Psychol.*, *69*(1), 105–129. https://doi.org/10.1146/annurev-psych-010416-044232

Wolfe, B. A., Kosovicheva, A. A., Leib, A. Y., Wood, K., & Whitney, D. (2015). Foveal input is not required for perception of crowd facial expression. *Journal of Vision*, *15*(4), 11–11.

Yang, Y., Tokita, M., & Ishiguchi, A. (2018). Is there a common summary statistical process for representing the mean and variance? A study using illustrations of familiar items. *I-Perception*, *9*(1), 2041669517747297.

Yörük, H., & Boduroglu, A. (2020). Feature-specificity in visual statistical summary processing. *Attention, Perception, & Psychophysics*, *82*, 852–864. https://doi.org/2:852–864 https://doi.org/10.3758/s13414-019-01942-x